

Here is the Hardware description of my Arduino Nano based mini ARCADE project that I built to teach young kids the basics of Arduino programming.

- * 1.3" monochrome SH1106 128x64 pixel OLED screen with I2C connections to SDA and SCL of Nano

- * Potentiometer with it's wiper pin connected to Nano pin A0

- * LDR with pull down resistor connected to Nano pin A1

- * RIGHT BTN connected to Nano pin D2 and with pull down resistor

- * Buzzer connected between +5V and collector of a NPN transistor. The Base of the transistor is connected to Nano pin D3

- * LEFT BTN and with pull down resistor connected to Nano pin D4

- * Haptic Motor connected to Nano pin D5

- * Ultrasonic Sensor with Echo connected to Nano pin D6 and Trigger connected to Nano pin D7

Please suggest some simple games that can be coded for playing on this hardware

ChatGPT said:

That's an **excellent little hardware setup** — your Arduino Nano-based **mini ARCADE** has just enough I/O and sensors to support a surprisingly wide range of interactive games.

It's perfect for teaching **sensor integration**, **logic**, and **simple graphics** using the SH1106 OLED.

Below is a curated list of **simple game ideas** designed for **young learners**, grouped by **complexity** and **which sensors/buttons** they use.

LEVEL 1: Easy Starter Games

(Use basic buttons, buzzer, and display — ideal for beginners.)

1. Button Duel (Reflex Game)

- **Goal:** Two players press their button (LEFT or RIGHT) as soon as a “GO!” signal appears on the OLED.
 - **Gameplay:**
 - A random delay, then “GO!” is displayed.
 - The first player to press wins.
 - Buzzer or haptic motor gives feedback for the winner.
 - **Teaches:** Debouncing, timing (`millis()`), random delays.
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2. Reaction Timer

- **Goal:** Press a button as quickly as possible when a visual signal (or buzzer) appears.
- **Hardware Used:** LEFT button, buzzer, OLED.
- **Feedback:** Display your reaction time in milliseconds.
- **Teaches:** Input timing, using `millis()` instead of `delay()`.

3. Catch the Dot

- **Goal:** Move a paddle left/right to “catch” a falling dot.
 - **Hardware Used:** Potentiometer for paddle position, OLED for animation, buzzer for success/fail.
 - **Gameplay:**
 - A dot falls from the top.
 - Move paddle with the potentiometer.
 - If the dot hits the paddle → score++.
 - If it misses → buzzer beeps and game ends.
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LEVEL 2: Sensor-Based Games

(Uses analog inputs like the LDR, ultrasonic, or potentiometer.)

4. Shadow Dodge (LDR Game)

- **Goal:** Control a character using **light intensity**.
 - **Gameplay:**
 - The player moves up/down depending on how much light hits the LDR.
 - Obstacles scroll from the right.
 - Move into darker/lighter zones to dodge them.
 - **Teaches:** Analog reading, thresholds, mapping sensor data.
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5. Ping Pong (Ultrasonic Paddle)

- **Goal:** Bounce a moving ball using an ultrasonic-controlled paddle.
 - **Hardware Used:** Ultrasonic sensor to move paddle up/down.
 - **Gameplay:**
 - Paddle moves based on distance.
 - Ball bounces off paddle and screen edges.
 - Miss → lose a life, buzzer beeps.
 - **Teaches:** Sensor reading, mapping, motion logic.
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6. Whack-a-Block

- **Goal:** Hit targets that appear randomly on screen using the potentiometer (move) and a button (hit).
 - **Hardware Used:** Potentiometer, LEFT button, buzzer.
 - **Gameplay:**
 - Blocks appear randomly.
 - Move your “hammer” horizontally and press to hit.
 - Buzz/haptic feedback when successful.
 - **Teaches:** Input coordination, random positions.
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LEVEL 3: Classic Mini-Games

(Combine multiple inputs and small logic loops.)

7. Snake (Potentiometer or LDR Controlled)

- **Goal:** Eat food and grow while avoiding self-collision.
 - **Controls:**
 - Potentiometer: left/right turning.
 - LDR: Light direction changes turn.
 - **Feedback:** Game over tone (3 short beeps).
 - **Teaches:** Arrays, grid-based movement, collision detection.
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8. Space Invaders (Potentiometer Movement + Buttons to Shoot)

- **Goal:** Move horizontally with the potentiometer and shoot descending aliens using LEFT or RIGHT button.
 - **Feedback:** Buzzer and haptic for explosions.
 - **Teaches:** Object tracking, simple sprite animation, shooting mechanics.
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9. Obstacle Runner (Ultrasonic Jump Game)

- **Goal:** Avoid oncoming obstacles by “jumping” — you raise your hand above the ultrasonic sensor to make the character jump.
- **Hardware Used:** Ultrasonic sensor, OLED, buzzer.
- **Gameplay:** Similar to Google Chrome’s Dino game.
- **Teaches:** Distance measurement, reaction timing, animation.



LEVEL 4: Creative Learning Projects

(Encourage kids to design their own logic or interaction.)

10. Mood Meter (LDR + Potentiometer)

- **Goal:** Create a “mood face” that changes expression based on ambient light and potentiometer position.
 - **Light** → **brightness of the smile.**
 - **Knob** → **switch between happy/sad/neutral.**
 - **Teaches:** Analog sensor mapping and conditional display logic.
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11. Simon Says (Memory Game)

- **Goal:** Repeat the sequence of beeps/flashes shown on screen.
 - **Hardware Used:** OLED for cues, buzzer for tones, LEFT/RIGHT buttons for input.
 - **Teaches:** Arrays, pattern recall, sequence validation.
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12. Balance Game (Potentiometer)

- **Goal:** Keep a small ball balanced at the center of a line using the potentiometer.
 - **Gameplay:**
 - Ball drifts left or right automatically.
 - Adjust potentiometer to keep it centered.
 - Buzzer or haptic buzzes if it drifts too far.
 - **Teaches:** PID concepts, analog control, calibration.
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Bonus Ideas:

- **Music Box:** Use potentiometer to select tones, buttons to play.
 - **Light Tracker:** LDR-based game — “chase” the right light range.
 - **Score Counter:** Simple app where button presses add/subtract points and display total.
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Recommendation for Classroom Teaching:

You can build a **menu-driven “ARCADE launcher”** that lists all these games:

1. Pong
2. Snake
3. Catch the Dot
4. Shadow Dodge
5. Space Invaders

- ➡ Navigate with potentiometer,
- ➡ Press LEFT to start,
- ➡ Press RIGHT to exit to menu.

That way, kids can try **different sensor-driven games** without re-uploading sketches.

PLAN

we'll create **5 beginner-friendly games** for your Arduino Nano-based mini ARCADE that progressively introduce key programming and hardware concepts — from digital inputs to sensor mapping and simple animation.

Each game will be:

- ✓ Fully commented (educational style).
- ✓ Simple enough for kids to follow.
- ✓ Designed for your exact hardware (OLED, potentiometer, ultrasonic, buttons, haptic motor, buzzer).
- ✓ Consistent layout and code structure.



5 GAMES FOR YOUR MINI ARCADE

#	Game	Key Hardware Used	Concept Taught
1	Button Duel (Reaction Game)	LEFT + RIGHT buttons, buzzer, OLED	Timing & logic
2	Catch the Dot	Potentiometer, buzzer, OLED	Analog mapping & collision detection
3	Shadow Dodge	LDR, buzzer, OLED	Using light sensor as control
4	Ultrasonic Pong	Ultrasonic sensor, buzzer, OLED	Distance mapping, animation
5	Space Invaders Mini	Potentiometer + buttons	Object motion & shooting

Below is **Game #1** — we'll start here, and then I'll generate the rest (2-5) in sequence if you confirm the format looks right.



Concepts Introduced:

- Digital input handling
- Random delay generation (`random()`)
- Simple game states (ready → play → result)
- Output feedback using buzzer/haptic motor

- ✓ Added a “**Starting...**” **animation** after LEFT BTN is pressed (3 dots appear sequentially).
- ✓ Made the game **automatically restart** after it’s over — either by pressing **LEFT BTN** to start a new round or both buttons together to return to the intro screen.
- ✓ Cleaned up visuals and comments for readability.

🧩 NEW FEATURES

- ✨ “**Starting...**” **animation** after LEFT BTN pressed — gives a short suspense buildup.
- 🔄 **Restart anytime** — press LEFT BTN to replay or both buttons to reset.
- 🎵 Clean feedback cycle — short buzz + haptic twice for winner.



MINI ARCADE GAME #2 — CATCH THE DOT



Concept:

A small dot moves left and right across the OLED screen.

The player uses the **potentiometer** to move their paddle at the bottom.

If the paddle catches the dot — they score a point!

If they miss, they lose a point.

The first to reach **+5 or -3 points** ends the game.

After the game, press **LEFT BTN** to restart.



Hardware (Arduino Nano Based)

- **SH1106 128x64 OLED (I2C)** → SDA = A4, SCL = A5
 - **Potentiometer** → Wiper → A0
 - **LEFT BTN (Restart)** → D4 (with pull-down resistor)
 - **Buzzer** → D3 → NPN transistor → +5V
 - **Haptic Motor** → D5 → transistor driver → +5V
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Teaching Focus

- Analog input → digital movement (potentiometer control)
- Simple collision detection
- Using the display for animation
- Feedback using buzzer + haptic



How It Works

- The **dot** falls vertically and resets after each round.

- The **paddle** position updates smoothly using a **low-pass filter** on the potentiometer reading.
- The **buzzer + haptic motor** give instant physical feedback.
- The **score** shows in the corner, and the **game restarts** with one button press.



MINI ARCADE GAME #3 — AVOID THE OBSTACLES



Concept:

You control a small **player block** that can move **left and right** using the **potentiometer**.

Falling **obstacles (blocks)** drop from the top — your goal is to **avoid getting hit**.

Each successful dodge gives you a point.

If you collide, you lose a life.

Lose 3 lives, and the game ends.

You can press **LEFT BTN** to start a new game after “Game Over.”



Hardware

Same as your current Mini Arcade:

- **OLED (SH1106)** → I2C
 - **Potentiometer (A0)** → controls player position
 - **LEFT BTN (D4)** → Start / Restart
 - **Buzzer (D3)** → via transistor
 - **Haptic Motor (D5)** → via transistor
 - (No need for Ultrasonic / LDR in this game)
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Learning Goals

- Continuous animation using timed motion
- Collision detection between rectangles
- Randomized spawning and progressive difficulty
- Sound + haptic feedback for collision
- Restart logic



How it works

- **Potentiometer** controls player movement smoothly.
- **Random blocks** fall from the top; you gain a point for dodging.

- **Collision detection** checks overlap between block and paddle.
- **Lives** decrease when hit; when all are gone, the game ends.
- **LEFT BTN** restarts the game.
- **Buzzer + haptic** provide physical feedback on crashes.

MINI ARCADE GAME #4 — SPACE SHOOTER

Game Concept

You control a **spaceship** at the bottom of the screen using the **potentiometer** to move left and right. Press **LEFT BTN** to **shoot lasers** at descending enemies.

Each hit gives you **1 point**.

If an enemy reaches the bottom, you lose **1 life**.

Lose all 3 lives → **Game Over**.

Press **RIGHT BTN** to **restart** the game.

Hardware (Arduino Nano)

Component	Connection
OLED Display (SH1106, 1.3")	I ² C (SDA=A4, SCL=A5)
Potentiometer	A0
LEFT BTN (Fire)	D4 (with pulldown)
RIGHT BTN (Restart)	D2 (with pulldown)
Buzzer (via NPN transistor)	D3
Haptic Motor	D5
Ultrasonic / LDR	Not used in this game

Learning Focus

- Motion and shooting logic
- Collision detection
- Multiple objects (bullet & enemies)
- Score and life tracking
- Game restart flow
- Visual feedback with simple animation

Features for Kids to Learn

- **Random motion + timing** = real-world animation logic

- **Collision detection** = bounding box math
- **Sound + haptic feedback** = tangible output
- **Game over logic + restart** = event-driven flow
- **Gradual difficulty** keeps it engaging



Key Improvements

- ✓ **Multiple bullets** (up to 5 on screen)
- ✓ **Faster firing** (150 ms cooldown)
- ✓ **Simple logic** — easy for kids to read and modify
- ✓ **No rapid repeat issues** — still controlled, but satisfying